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Buddha Institute of Technology GIDA, Gorakhpur

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" Comprehensive Analysis of Water Quality of Rapti River in Gorakhpur Region"

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Introduction

Background:

- In Gorakhpur, the Rapti River plays a dominant role in shaping the city's geography, hydrology, and ecology.
- The region lies in the middle Gangetic alluvial plains, which are fertile but highly flood-prone due to the river's behavior.

Importance:

- Supports aquatic ecosystems and biodiversity
- Provides irrigation for large agricultural areas
- Ensures safety and stability of excavation projects.

Context:

- Water Quality Concerns.
- Flood and Erosion Issues.
- Urbanization and Population Pressure

Problem Statement

Define the Problem:

- Water Pollution.
- Frequent Flooding.
- Riverbank Erosion and Siltation.

Why it matters:

- Environmental Impact.
- Public Health Concerns.
- Agricultural Dependency

Current Gap:

- Lack of Integrated Studies.
- Limited Pollution Source Identification.
- Inadequate Use of Modern Techniques.

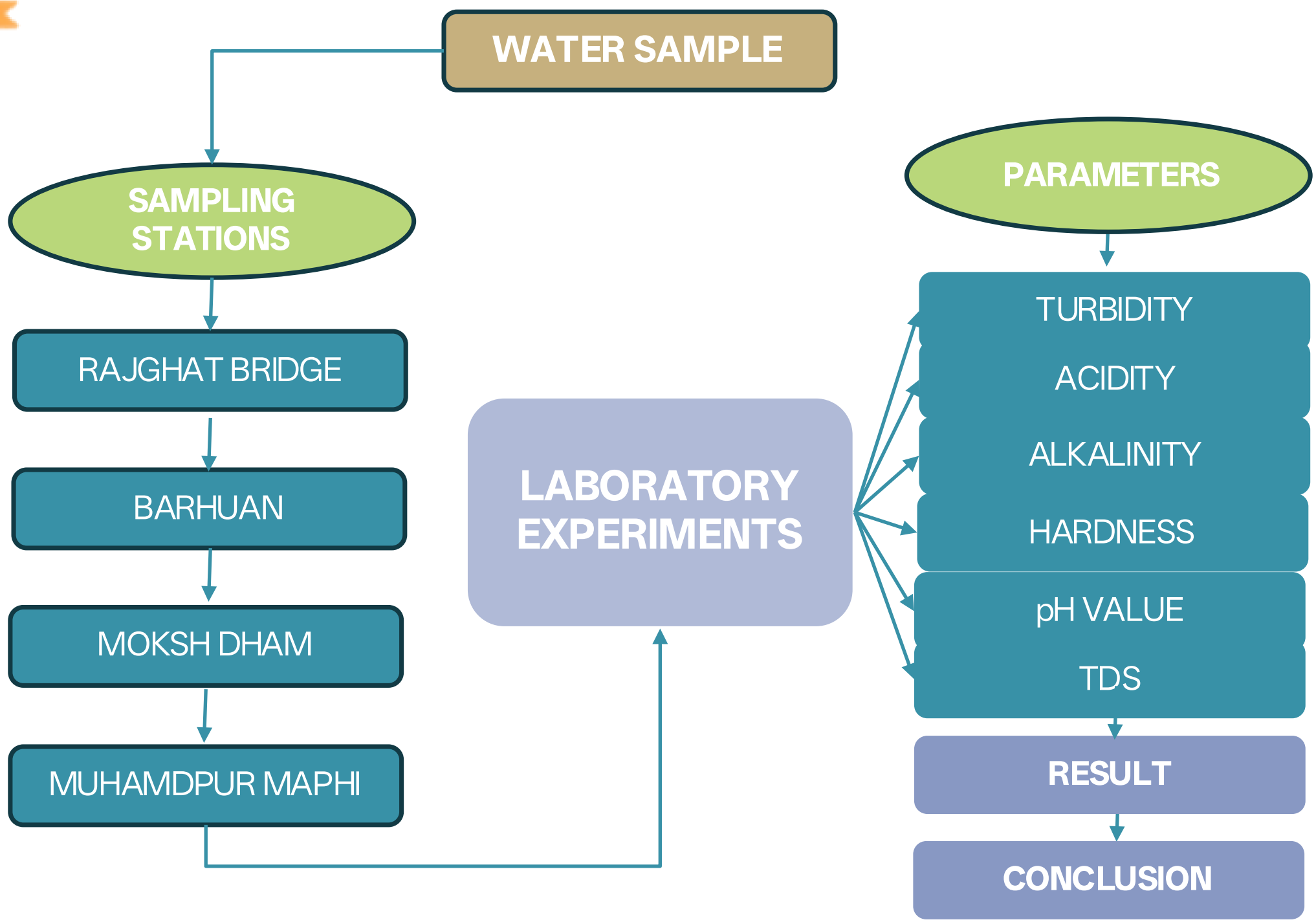
Objectives

- **Objective 1:** To access variation in water quality characteristics across different location in Gorakhpur .
- **Objective 2:** Evaluate the presence and concentration of pollutants .
- **Objective 3:** Identify the sources of pollution (domestic, industrial, agricultural runoff, etc.).

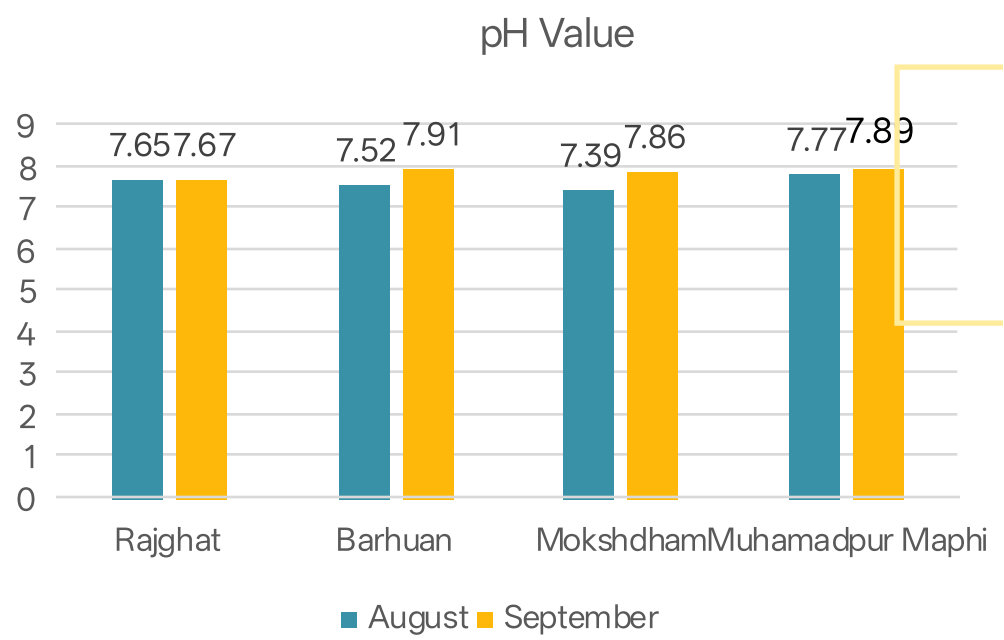
Literature Survey

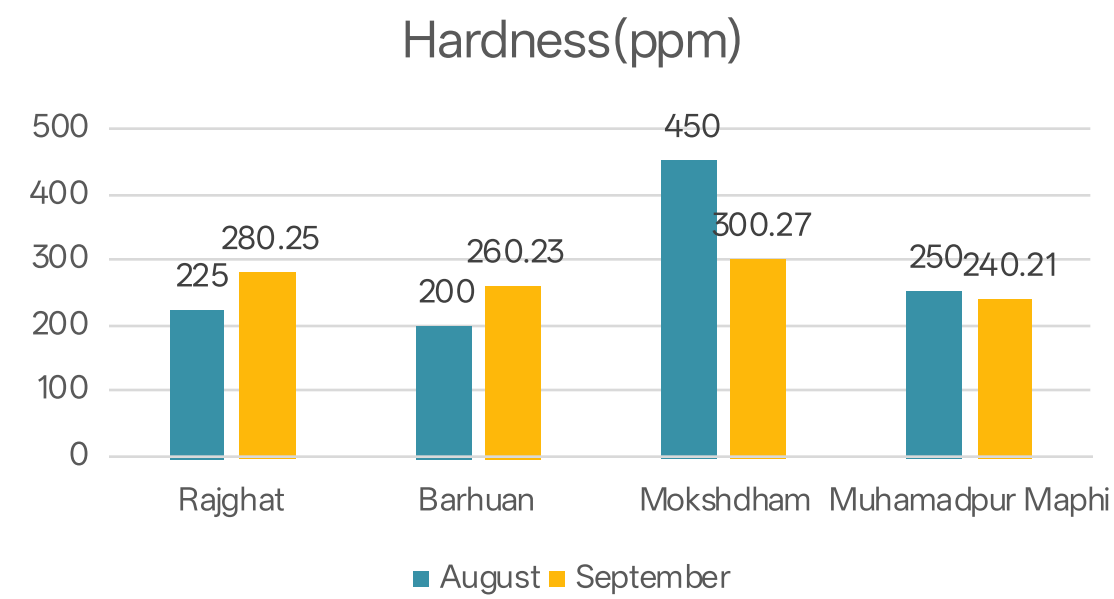
- **Yadav, D. & Pandey, G. (2021)-** Gorakhpur is situated from bank of Rapti River. Rapti River is joined at Sohgaura with Ami River which is highly polluted as compare to Rapti River. Rapti River has a confluence into Ghaghra River at Kaparwar Ghat. The present study is conducted to determine the water quality of Rapti River using physical and Chemical parameters including pH, Turbidity, Total dissolved solids, Dissolved Oxygen (DO), and Total Hardness by collecting Three sets of samples from five locations from December, 2020 to February, 2021.
- **Mishra, KD (2020)-** This paper illustrates the quality of Yamuna River water from the Agra region of Uttar Pradesh. This river is the second largest tributary of Ganga which flows from Uttrakhand to Haryana and then enters into Noida, Uttar Pradesh, and lastly merges into Ganga River at Triveni Sangam in Prayagraj, Uttar Pradesh.
- **Gond, M. & Pandey, G. (2017**)- In this study, water quality evaluation was completed in 8 stations of Gorakhpur city. The study's point was to survey the physico-chemical parameters of drinking water quality from groundwater sources. It is found that the increment in the level of a portion of the physico chemical parameters of ground water has risen as a vital issue in urban areas.

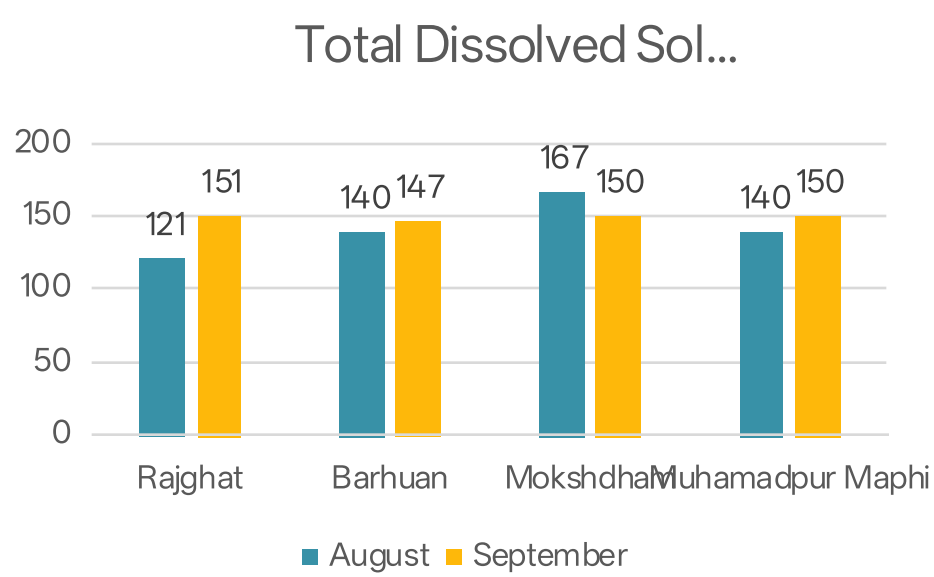
Proposed Solution / Methodology

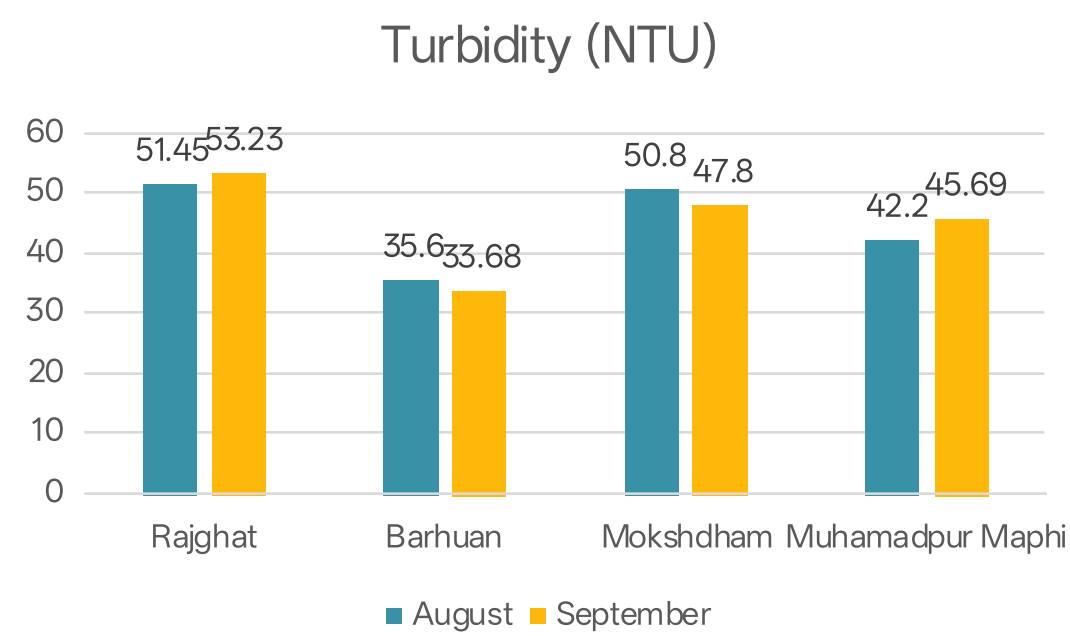


Results & Analysis









Conclusion & Future Work

- The assessment of water quality parameters in the Rapti River revealed significant variations in physical, chemical, and biological characteristics.
- The pH, hardness, alkalinity, and chloride content were found to be within the acceptable limits for drinking water, as per the Bureau of Indian Standards.
- The assessment of water quality parameters of the Rapti River provides essential insights into the current status of river health and the potential impacts of natural and human-induced activities on this important freshwater resource.
- In conclusion, this study emphasizes the importance of regular water quality monitoring, the need for sustainable water resource management, and the urgent requirement for pollution mitigation to ensure that the Rapti River remains a viable and safe water source for future generation.

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Institution:

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Thank You